

1. Administrative & Study Identification

Study Title: A Study to Evaluate the Efficacy and Safety of Factor IX Gene Therapy With PF-06838435 in Adult Males With Moderately Severe to Severe Hemophilia B **Official Title:** Phase 3, Open-label, Single-arm Study to Evaluate Efficacy and Safety of FIX Gene Transfer With PF-06838435 (rAAV-Spark100-hFIX-R338L) in Adult Male Participants With Moderately Severe to Severe Hemophilia B (FIX:C ≤2%) (BeneGene-2) **Short Title/Acronym:** BENEGENE-2 **Protocol Number:** C0371002 **NCT Number:** NCT03861273 **Posting ID:** 618752104271303894 **Sponsor/Company Name:** Pfizer **Investigational Product:** PF-06838435 (fidanacogene elaparvovec / BEQVEZ™) **Drug Preferred Name:** Fidanacogene elaparvovec **Phase of Development:** Phase III **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the annualized bleeding rate (ABR) for participants treated with gene therapy (fidanacogene elaparvovec) versus standard of care (SOC) therapy (routine FIX prophylaxis replacement regimen). **Secondary Objectives:** To evaluate vector-derived Factor IX (FIX:C) levels, the annualized infusion rate (AIR) of exogenous Factor IX activity, and annualized Factor IX activity consumption.

2.2 Background & Rationale To address the significant treatment burden in adults with moderately severe to severe Hemophilia B. Standard of care requires lifelong, regular intravenous infusions of FIX to prevent bleeding episodes. PF-06838435 is a one-time adeno-associated virus (AAV) gene therapy designed to introduce genetic material into the cells to compensate for missing or non-functioning Factor IX, offering the potential for stable FIX expression and a reduction or elimination of routine prophylaxis.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Intra-patient historical control utilizing data from lead-in study C0371004) **Blinding:** Open-label **Randomization:** N/A **Criteria Type:** PHASE_REQUIREMENT (Criteria Text: Trial must be in PHASE3)

3.2 Planned Enrollment & Duration Targeted Enrollment: 51 enrolled (45 treated) **Number of Sites:** Multinational (Sites in

the US, Japan, UK, Canada, France, Germany, etc.) **Estimated Study Start:** 29-Jul-2019 **Estimated Primary Completion:** 16-Nov-2022

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult males, 18 to 65 years of age. 2. Confirmed diagnosis of moderately severe to severe Hemophilia B (endogenous Factor IX circulating activity $\leq 2\%$). 3. Completed a minimum of 6 months of routine FIX prophylaxis therapy during the lead-in study (C0371004) with $\geq 50\%$ prior exposure days to any FIX protein products.

4.2 Exclusion Criteria 1. History of FIX inhibitors. 2. Presence of pre-existing anti-AAV-Spark100 neutralizing antibodies. 3. Clinically significant liver disease or other unstable medical conditions.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose of PF-06838435 (fidanacogene elaparovector). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** One-time infusion evaluated over an initial course of 6 years (with a planned long-term follow-up of 15 years).

5.2 Concomitant & Prohibited Therapy Permitted: Corticosteroids/glucocorticoids for the management of immune-mediated transaminase elevations and/or decreased FIX levels. **Prohibited:** Routine exogenous FIX prophylaxis post-infusion (unless medically necessitated by targeted bleeding episodes or surgical events).

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures
Screening	Pre-infusion	Informed Consent, Lead-in Study (C0371004) Data Collection, Neutralizing Antibody Testing
Baseline	Day 0	Single IV Infusion of Gene Therapy, Vitals
Interim	Weeks 1-52	Frequent Safety Labs (ALT/AST monitoring), Corticosteroid initiation if needed, FIX:C levels, ABR tracking
End of Study	Year 6	Final Efficacy Assessment, Transition to Long-term Follow-up Registry (15 years total)

7. Outcome Measures (Endpoints)

7.1 Primary Outcome Definition: Annualized Bleeding Rate (ABR) for all bleeding episodes. Measured to evaluate non-inferiority and superiority from Week 12 to Month 15 post-infusion compared

to the lead-in standard of care prophylaxis period. **Measurement Method:** Patient reporting and clinical evaluation to count total bleeding episodes, excluding surgical procedures. Calculation: (Number of total bleeding episodes) * 365.25 / (Date of last day - date of first day + 1).

7.2 Secondary & Exploratory Outcomes - ABR for Treated Bleeds From Week 12 to Month 15. - Annualized Infusion Rate (AIR) of exogenous Factor IX activity From Week 12 to Month 15. - Steady State Circulating Factor IX (FIX:C) From Week 12 to Month 15.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Liver function testing (aminotransferase levels) to monitor for AAV-mediated hepatotoxicity, and testing for development of FIX inhibitors. **Serious Adverse Events (SAEs):** Evaluated continuously for events such as thrombosis, infusion-related reactions, and malignancies. **Data Monitoring Committee (DMC):** Internal and external safety review committees governing advanced therapies and regenerative medicines.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population (all subjects receiving the gene therapy infusion). **Sample Size Justification:** Sized to statistically demonstrate non-inferiority and superiority in reducing ABR compared to the lead-in study baseline. **Handling of Missing Data:** Standardized calculation of ABR with precise inclusion/exclusion timeframes.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite safety monitoring during the initial 12-week post-infusion window, followed by extended interval reporting. **Audit Trail:** eCRF locking and stringent regulatory reporting requisite for FDA/EMA gene therapy track evaluations.

11. Study Identifiers & Governance

Primary ID: C0371002 **Registry Number:** NCT03861273 **Posting ID:** 618752104271303894 **Sponsor/Lead Organization:** Pfizer **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Study Title:** A Study to Evaluate the Efficacy and

12. Clinical Framework (Hemophilia B Specific)

Disease Area / Preferred Name: Hemophilia B **Preferred UMLS Name:** Gene Therapy for Hemophilia B **Semantic Type:** Disease or Syndrome
MeSH Code: D002836 **SNOMED ID:** 41788008 **Definition:** A deficiency of blood coagulation factor IX inherited as an X-linked disorder. (Also known as Christmas Disease, after the first patient studied in detail, not the holy day.) Historical and clinical features resemble those in classic hemophilia (HEMOPHILIA A), but patients present with fewer symptoms. Severity of bleeding is usually similar in members of a single family. Many patients are asymptomatic until the hemostatic system is stressed by surgery or trauma. Treatment is similar to that for hemophilia A.
Diagnosis Threshold: Endogenous FIX:C $\leq 2\%$ **Medical Methodology:** AAV-Mediated Gene Therapy (fidanacogene elaparovvec)
Optimization Target: Sustained vector-derived FIX expression and reduction/elimination of ABR

13. Study Procedures & Methodology

Management Pathway: AAV Vector infusion with close hepatic monitoring **Education Protocol (HCP):** Transaminase elevation management and reactive corticosteroid dosing **Education Protocol (Patient):** Bleeding event reporting and adherence to follow-up testing **Quality Audit Mechanism:** Central laboratory monitoring of FIX activity and immune responses

14. Observation & Follow-Up Schedule

Total Duration: 6 Years (followed by 9-year extension for 15 years total) **Onsite Visit Cadence:** Frequent evaluations during Months 1-3, followed by bi-annual/annual assessments **Remote Monitoring Frequency:** Continuous patient-reported bleeding diary tracking **Checklist Review Interval:** Regular clinical checks for exogenous FIX use

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				